TOPFAITH UNIVERSITY
MKPATAK, AKWA IBOM STATE, NIGERIA

B.Sc. DEGREE EXAMINATION

FACULTY: COMPUTING AND APPLIED SCIENCES
DEPARTMENT: PHYSICS
COURSE CODE: PHY 107
COURSE TITLE: GENERAL LAB. PHYSICS I (PRACTICAL)
INSTRUCTION: ANSWER ONE (1) QUESTION ONLY

SESSION: 2023/2024
SEMESTER: PRIMO
CREDIT UNIT: 1
TIME: 1 HOUR

1. Given that experiment variables T and m are related to some constants M_o and g as shown in the equation below,

$T = 2\pi\sqrt{\frac{M+M_o}{g}}$

(a) Transform this equation into linear form. (5 marks)
(b) Sketch and label the nature of your graph. (5 marks)
(c) (i) Describe how you will determine the constants g and M_o from a suitable graph. (4 marks)
(ii) What is the physical significance of M_o? (2 marks)
(d) In the following set of repeated readings 24.3, 24.8, 24.2, 24.4, 24.6, 24.1, 24.6, 24.7, 24.6, 24.7. Calculate the,
(i) mean (2 marks)
(ii) range (1 marks)
(iii) standard error of the mean. (3 marks)
(e) Sketch the nature of the graphs and name the following equations:
(i) $y = mx + c$ (2 marks)
(ii) $y = ax^2 + bx + c$ (4 marks)
(iii) $y = \frac{c}{x}$ (2 mark)

[Total = 30 marks]

2. In an experiment to determine the relationship between the periodic time T and the length y of the part of an obstructed simple pendulum allowed to oscillate, the following results were obtained:

y (cm)	**Time for 20 oscillations**			**Period T_y (S)**	$T_y^2\ (S)^2$	$T^2 = T_L^2 - T_y^2\ (S)^2$
	t_1 (s)	t_2 (s)	t_{av} (s)			
40.00	25.52	25.40				
35.00	23.60	24.00				
30.00	22.60	22.20				
25.00	19.20	21.00				
20.00	18.40	18.20				
15.00	15.90	16.10				
10.00	12.00	14.00				

(a) Complete the table above (take T_L = 2.0 s). (12 marks)
(b) Plot a graph of $T_L^2 - T_y^2\ (S)^2$ against y (cm). (5 marks)
(c) Transform the equation below to a linear form: $T = 2\pi\sqrt{\frac{L-y}{g}}$ (8 marks)
(d) Determine the,

(i) slope of the graph (2 marks)

(ii) intercept on the y axis (1 mark)

(e) State two precautions you will take in carrying out the experiment. (2 marks)

[Total = 30 marks]

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COURSE COORDINATOR: DR ENOBONG J. ODUOBUK
COURSE LECTURER: DR. ENOBONG J. ODUOBUK

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MARKING GUIDE

QUESTION No. 1

(a) $T = 2\pi\sqrt{\frac{M+M_o}{g}}$

Square both side of the equation to remove the square root sign,

$T^2 = 4\pi^2\ (\frac{M+M_o}{g})$ (1 mark)

$gT^2 = 4\pi^2\ (M + M_o)$ (1 mark)

$\frac{gT^2}{4\pi^2} = M + M_o$ (1 mark)

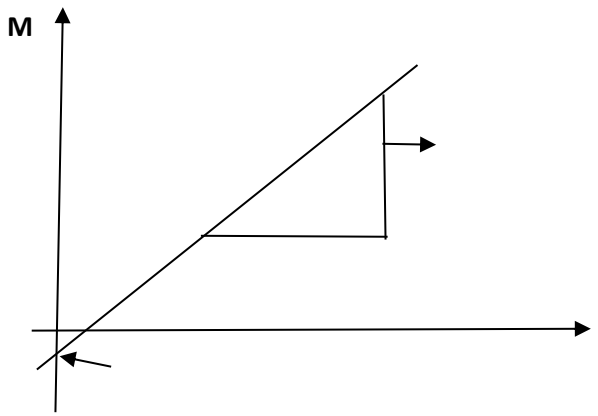
Making M the subject

$M = \frac{gT^2}{4\pi^2} - M_o$ (1 mark)

Therefore,

$M = \frac{g}{4\pi^2} . T^2 - M_o$ (1 mark)

(b) The equation, $M = \frac{g}{4\pi^2} . T^2 - M_o$ is a straight-line graph with positive slope of $\frac{g}{4\pi^2}$ and a negative intercept of M_o. (2 marks)



(3 marks)

(c) (i) M_o and g can be determined from the graph as follows;

$\frac{g}{4\pi^2}$ = slope of the graph ($1\frac{1}{2}$ marks)

g = $4\pi^2$ x slope of the graph ($1\frac{1}{2}$ marks)

M_o = intercept on the y-axis. (1 mark)

(ii) The significance of M_o is initial mass. (2 marks)

(d) (i) mean $\underline{X} = \frac{24.3+24.8+24.2+24.4+24.6+24.1+24.6+24.7+24.6+24.7}{10}$ (1 mark)

$\underline{X} = \frac{245}{6}$ = 24.5 (1 mark)

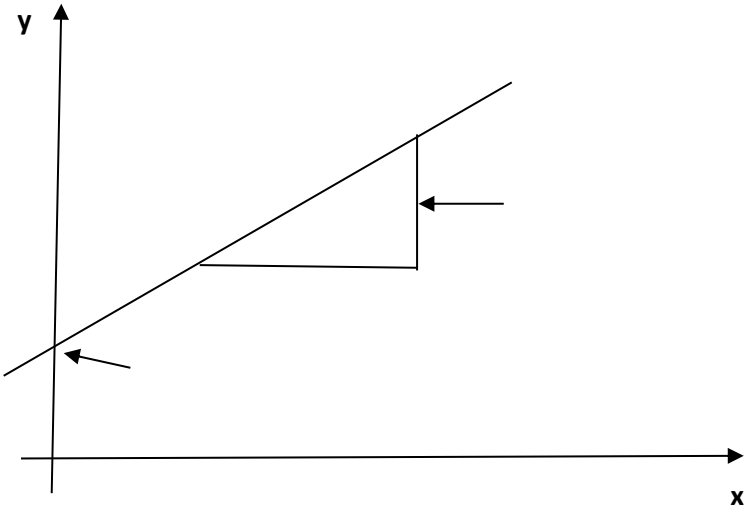
(ii) The range = 24.8 – 24.1 = 0.7 (1 mark)

(iii) Standard error of the mean, Se

Se = $\frac{w}{n} = \frac{24.8-24.1}{10}$ (2 marks)

Se = $\frac{0.7}{6}$ = 0.05 (1 mark)

(e) (i) y = mx + c: This is straight line equation. ($\frac{1}{2}$ mark)



($1\frac{1}{2}$ marks)

(ii) y = $ax^2 + bx + c$: Parabola Equation ($\frac{1}{2}$ mark)

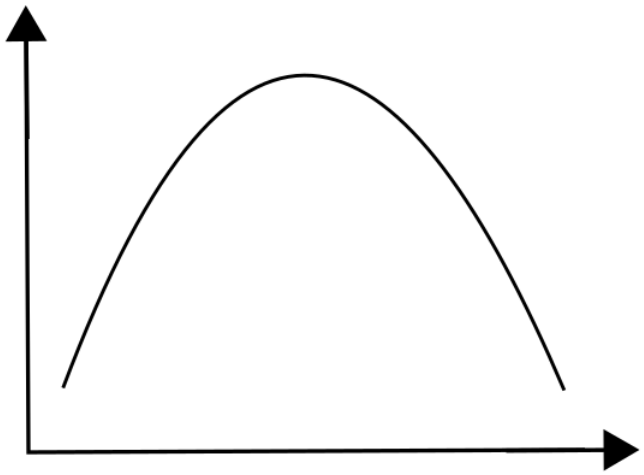
y – c = ax^2 ($\frac{1}{2}$ mark)

$\frac{y-c}{x} = \frac{x(ax+b)}{x}$ ($\frac{1}{2}$ mark)

$\frac{y-c}{x}$ = ax + b ($\frac{1}{2}$ mark)

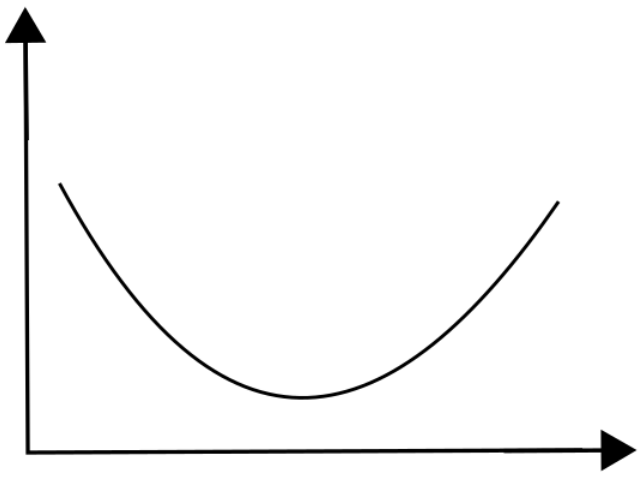
The graph of $\frac{y-c}{x}$ against x ($\frac{1}{2}$ mark)

a = slope, b = intercept ($\frac{1}{2}$ mark)



(1 mark)

(iii) y = $\frac{c}{x}$: Hyperbola equation ($\frac{1}{2}$ marks)



($1\frac{1}{2}$ marks)

[Total = 30 marks]

QUESTION No. 2

(a)

y (cm)	**Time for 20 oscillations**		*(3 marks)*	**Period T_y (S)** *(3 marks)*	$T_y^2\ (S)^2$ *(3 marks)*	$T^2 = T_L^2 - T_y^2\ (S)^2$ *(3 marks)*
	t_1 (s)	t_2 (s)	t_{av} (s)			
40.00	25.52	25.40	25.46	1.273	1.62	2.38
35.00	23.60	24.00	23.80	1.190	1.42	2.58
30.00	22.60	22.20	22.30	1.115	1.24	2.76
25.00	19.20	21.00	20.10	1.005	1.01	2.99
20.00	18.40	18.20	18.30	0.915	0.84	3.16
15.00	15.90	16.10	16.00	0.800	0.64	3.36
10.00	12.00	14.00	13.00	0.650	0.42	3.58

(12marks)

(b) **(See attached below)**

- ❖ Title (½ mark)
- ❖ Scale (½ mark)
- ❖ Labelled axes (y – axis and x-axis) (1 mark)
- ❖ Plotted points, line of best fit and slope (3 marks)

(c) T = $2\pi\sqrt{\frac{L-y}{g}}$

Squaring both sides to remove the square root sign (1 mark)

$(T)^2 = (2\pi\sqrt{\frac{L-y}{g}})^2$ (1 mark)

$T^2 = 4\pi^2(\frac{L-y}{g})$ (1 mark)

$T^2 = \frac{4\pi^2 L}{g} - \frac{4\pi^2 y}{g}$ (1 mark)

$T^2 = -\frac{4\pi^2 y}{g} + \frac{4\pi^2 L}{g}$ (1 mark)

Slope of the graph = $-\frac{4\pi^2 y}{g}$ ($1\frac{1}{2}$ marks)

Intercept on T^2 axis = $\frac{4\pi^2 L}{g}$ = c ($1\frac{1}{2}$ marks)

(d) (i) Slope of the graph = $\frac{\Delta T^2}{\Delta y} = \frac{3.80-2.50}{36.50-5.00}$ (1 mark)

$= \frac{1.30}{31.50} = 0.04$ (1 mark)

(ii) Intercept = 4.0 seconds (1 mark)

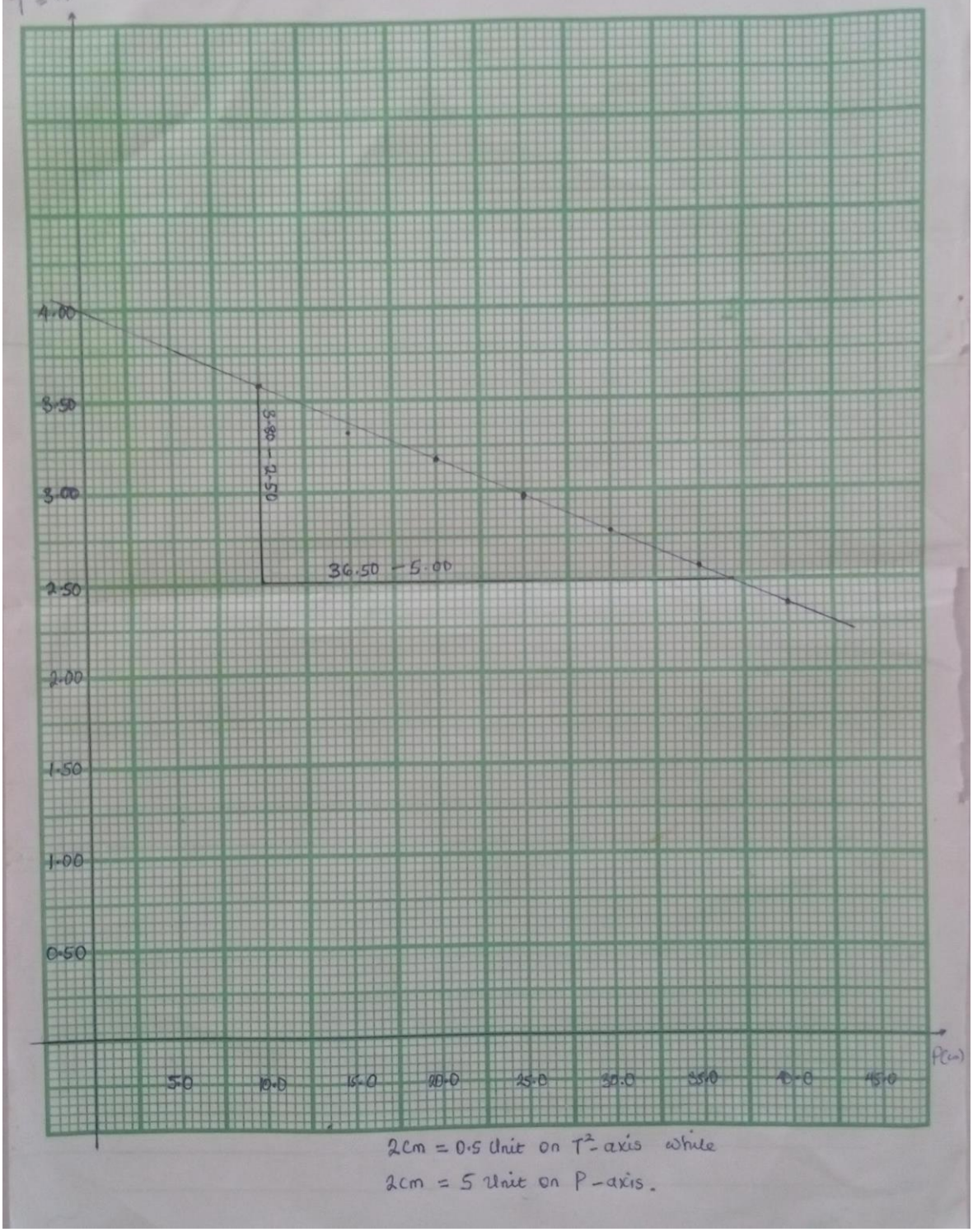
(e) Precautions (Any two; *1 mark for each*) (2 marks)

- ❖ I avoided parallax error when taking readings on the meter rule.
- ❖ I ensured repeated counting in order to avoid random error.
- ❖ I ensured zero error on the stop watch is corrected.
- ❖ I ensured a relatively small angle of oscillations throughout the experiments.

[Total = 30 marks]

$T^2 = \bar{I}_v - \bar{I}_p \ (s^2)$

The Graph of T^2 against P.



2Cm = 0.5 Unit on T^2 axis while

2Cm = 5 Unit on P-axis.